A FINITE TYPE MAP WITH NONEMPTY GENERIC FIBRE IS SURJECTIVE OVER A NONEMPTY OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6509-6515 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6517-6521, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion lemma-empty-generic-fibre (lines 6495-6507) has a pure-pointer proof and is not collected.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The algebra description of the image of a finite type ring map (Algebra, lemma-characterize-image-finite-type); cited input with the affine-open reduction written in the retained span.

Source reference. Original lemma 00FH: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6521–6521. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a finite type morphism of schemes. Assume Y irreducible with generic point η. If $X_\eta \neq \emptyset$ then there exists a nonempty open $V \subset Y$ such that $X_V = V \times_Y X \to V$ is surjective.*

Proof. This follows, upon taking affine opens, from Algebra, Lemma 00FH. (Of course it also follows from generic flatness.) □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.